



Differential Geometry

微分几何 - 学习笔记

Alpha Version

潜龙勿用 容与

本笔记根据上海交通大学 Tobias Diez 老师的 Differential Geometry 课程内容编写

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1 曲线论

1.1 参数化曲线 (Parameterized Curve)

1.1.1 定义 1: 曲线 (Curve)

设 I 是一个区间, $\mathbb{R}^n$ 为 n 维 Euclidean 空间 (通常 $n = 2, 3$), c 为一个光滑的映射,

$$c : I \longrightarrow \mathbb{R}^n, t \longmapsto c(t) \quad (1.1.1)$$

则称 c 为**参数化曲线**. 称其像的集合 $\{c(t) \mid t \in I\}$ 为**曲线**.

参数 t 不一定代表时间, 但可借用物理背景, 定义速度矢量 $\dot{c}(t) = \frac{dc}{dt}$ 与加速度矢量 $\ddot{c}(t) = \frac{d\dot{c}}{dt}$.

1.1.2 定义 2: 正规曲线 (Regular Curve)

设一条参数化曲线 c , 若 c 满足:

$$\dot{c}(t) \neq 0, \forall t \in I \quad (1.1.2)$$

则称 c 为一条**正规曲线**.

正规曲线没有尖点和角, 且每一点都有一个方向明确的非零切向量.

1.1.3 定义 3: 曲线的长度 (Length of a Curve)

设参数化曲线 $c : I \rightarrow \mathbb{R}^n$ ($I = [a, b]$ or (a, b)), c 的长度为:

$$L(c) := \int_a^b \|\dot{c}(t)\| dt \quad (1.1.3)$$

1.1.4 定义 4: 微分同胚 (Diffeomorphism)

设 φ 是一个映射, 若满足:

1. φ 是光滑的;
2. φ 是一个双射;
3. 其逆映射 φ^{-1} 也是光滑的;

则称 φ 为一个**微分同胚**.

1.1.5 定义 5: 重参数化 (Reparameterization)

设参数化曲线 $c_1 : I_1 \rightarrow \mathbb{R}^n$, 微分同胚 $\varphi : I_2 \rightarrow I_1$ ($\varphi'(s) > 0, \forall s \in I_2$), 则称另一参数化曲线 $c_2(s)$:

$$c_2 = c_1 \circ \varphi : I_2 \longrightarrow \mathbb{R}^n \quad \& \quad c_2(s) = c_1(\varphi(s)) \quad (1.1.4)$$

为 c_1 通过 φ 的重参数化.

1.1.6 命题 1

若 c_2 为 c_1 的一条重参数化曲线, 则 c_1 与 c_2 长度相同. 换言之, 曲线的长度与参数化的方式无关.

$$\begin{aligned} \dot{c}_2(s) &= \frac{dc_2}{ds}(s) \xrightarrow{\text{链式法则}} \frac{dc_1}{dt}(\varphi(s)) \cdot \varphi'(s) \\ \implies L(c_2) &= \int_{a_2}^{b_2} \left\| \frac{dc_2}{ds}(s) \right\| ds = \int_{a_2}^{b_2} \left\| \frac{dc_1}{dt}(\varphi(s)) \right\| \cdot \|\varphi'(s)\| ds = \int_{a_1}^{b_1} \left\| \frac{dc_1}{dt}(t) \right\| dt = L(c_1) \end{aligned}$$

1.1.7 命题 2

若 $\|\dot{c}(t)\| = 1$, 则有

$$\dot{c}(t)\ddot{c}(t) = 0 \quad (1.1.5)$$

$$\|\dot{c}(t)\| = 1 \implies \|\dot{c}(t)\|^2 = \dot{c}(t) \cdot \dot{c}(t) = 1 \xrightarrow{\text{求导}} \ddot{c}(t) \cdot \dot{c}(t) + \dot{c}(t) \cdot \ddot{c}(t) = 0 \implies \dot{c}(t)\ddot{c}(t) = 0$$

1.1.8 定义 6: 弧长参数化曲线 (Arclength Parameterized Curve)

设参数化曲线 $c : I \rightarrow \mathbb{R}^n$ 具有单位速率 (即 $\|\dot{c}(t)\| \equiv 1, \forall t \in I$), 则称其为弧长参数化的. 其长度为:

$$L(c_{[a,b]}) = \int_a^b \|\dot{c}(t)\| dt = b - a \quad (1.1.6)$$

1.1.9 命题 3

每条正规曲线都可被重参数化为一条弧长参数化曲线.

设微分同胚 $\varphi : I \rightarrow I$ ($s \mapsto t$) 使正规曲线 $c(t)$ 重参数化为弧长参数化曲线 $\tilde{c}(s)$,

$$s = \psi(t) = \int_{t_0}^t \|\dot{c}(t)\| dt = L(c_{[t_0,t]}), \quad t_0 \in I \quad \left(\frac{d\psi(t)}{dt} = \|\dot{c}(t)\| > 0 \right)$$

$$\psi^{-1} \equiv \varphi \implies \frac{d\varphi}{ds} = \frac{1}{\frac{d\psi}{dt}(\varphi(s))} = \frac{1}{\|\dot{c}(\varphi(s))\|}$$

$$\|\dot{c}(s)\| = \left\| \frac{dc}{dt}(\varphi(s)) \right\| \cdot \|\varphi'(s)\| = \left\| \frac{dc}{dt}(\varphi(s)) \right\| \cdot \frac{1}{\|\dot{c}(\varphi(s))\|} = 1$$

1.1.10 定理 1: 活动标架 (Moving Frame)

设 $c: I \rightarrow \mathbb{R}^n$ 为一条单位速度曲线, 且 $\|\ddot{c}\| \neq 0, \forall t \in I$, 其切向量 (Tangent Vector) 与法向量 (Normal Vector) 为:

$$T(t) := \dot{c}(t) \quad N(t) := \frac{\ddot{c}(t)}{\|\ddot{c}(t)\|} \quad (1.1.7)$$

$T(t)$ 与 $N(t)$ 均长度为 1 且相互垂直, 构成一对正交归一向量.

若 c 位于二维 Euclidean 空间中, 则 $T(t), N(t)$ 构成一个沿 c 移动的标准正交基, 称其为 c 的活动标架 (Moving Frame) 或称 Frenet-Serret 标架.

1.1.11 定义 7: 曲率 (Curvature)

设弧长参数化曲线 $c: I \rightarrow \mathbb{R}^n$, 有

$$\kappa(t) = \|\ddot{c}(t)\| \in [0, \infty) \quad (1.1.8)$$

称 $\kappa(t)$ 为 c 在 t 处的曲率.

1.1.12 非弧长参数化曲线的曲率

若正规曲线 c 是非弧长参数化的, 令 $\tilde{c} = c \circ \varphi$ 为其弧长参数化映射, $\tilde{\kappa}$ 为 $\tilde{c}$ 的曲率, 则 c 的曲率为:

$$\kappa(t) = \tilde{\kappa}(\varphi^{-1}(t)) \quad (1.1.9)$$

$$\kappa(t) = \frac{\sqrt{\|\ddot{c}(t)\|^2 - \dot{v}^2(t)}}{v^2(t)} = \frac{\sqrt{\|\ddot{c}(t)\|^2 - [\dot{c}(t) \cdot \ddot{c}(t)]^2}}{v^3(t)} \quad (1.1.10)$$

其中, 速率 $v(t) = \|\dot{c}(t)\|$.

$$ds = \|\dot{c}(t)\| dt \implies v(t) = \frac{ds}{dt}$$

$$T(s) = \frac{\dot{c}(t)}{ds/dt} = \frac{\dot{c}(t)}{v(t)} \implies \frac{dT}{ds} = \frac{dT/dt}{ds/dt} = \frac{1}{v(t)} \frac{dT}{dt} = \frac{\ddot{c}(t)v(t) - \dot{v}(t)\dot{c}(t)}{v^3(t)}$$

$$\dot{v}(t) = \frac{d}{dt}\|\dot{c}(t)\| = \frac{d}{dt}\sqrt{\dot{c}(t) \cdot \dot{c}(t)} = \frac{\dot{c}(t) \cdot \ddot{c}(t)}{v(t)} \implies \frac{dT}{ds} = \frac{\ddot{c}(t)v^2(t) - \dot{c}(t)[\dot{c}(t) \cdot \ddot{c}(t)]}{v^4(t)}$$

$$\kappa(t) = \left\| \frac{dT}{ds} \right\| = \frac{1}{v^4(t)} \sqrt{\{\ddot{c}(t)v^2(t) - \dot{c}(t) [\dot{c}(t) \cdot \ddot{c}(t)]\}^2} = \frac{\sqrt{\|\ddot{c}(t)\|^2 - [\dot{c}(t) \cdot \ddot{c}(t)]^2}}{v^3(t)}$$

$$\xrightarrow{\dot{c}(t) \cdot \ddot{c}(t) = \dot{v}(t)v(t)} \kappa(t) = \frac{\sqrt{\|\ddot{c}(t)\|^2 - \dot{v}^2(t)}}{v^2(t)}$$

1.2 曲线的局部性质与整体结构

1.2.1 定义 8: 回路 (Loop) & 简单回路 (Simple Loop)

设参数化曲线 $c: [a, b] \rightarrow \mathbb{R}^n$, 若 $c(a) = c(b)$, 且 c 在端点 a 和 b 处的所有单侧导数均相等, 即

$$\left. \frac{d^k}{dt^k} \right|_{t=a} c(t) = \left. \frac{d^k}{dt^k} \right|_{t=b} c(t), \quad \forall k \geq 1 \quad (1.2.1)$$

则称 c 为回路.

若 c 不与自身相交, 即

$$c(t_1) \neq c(t_2), \quad \forall t_1, t_2 \in (a, b) \text{ 且 } t_1 \neq t_2 \quad (1.2.2)$$

则称 c 为简单回路.

1.2.2 定理 2: Jordan 曲线定理

平面 $\mathbb{R}^2$ 上的每条简单回路, 都将平面分成两个集合: 一个是有界的, 称为回路的内部; 另一个是无界的, 称为回路的外部.

1.2.3 定理 3: 等周不等式 (Isoperimetric Inequality)

设简单回路 $c: [a, b] \rightarrow \mathbb{R}^2$, 长度为 L , 所围区域的面积为 A , 则有

$$4\pi A \leq L^2 \quad (1.2.3)$$

当且仅当 c 为一个圆周时, 等号成立.

1.2.4 引理

设一条简单正规回路 $c: I \rightarrow \mathbb{R}^2$, 则一定存在 $t_0 \in I$, 使得在 c 处的密切圆存在, 且在回路内部.

定义 p 点处的极大圆 (maximal circle) c_p : 与 c 相切于 p 且位于 c 内部的半径最大的圆, c_p 与 c 至少还有另一个交点 q . 择一参数 $t \in [a, b]$, 使得 $p = c(t)$, 那么

1. 若 c 在 p 处的密切圆位于 c 内部, 则符合上述引理;
2. 假设该密切圆不在 c 内部, 则 p 处极大圆的半径一定严格小于该密切圆, 密切圆作为曲线的二阶近似, 保证了在 p 的某一邻域内, 其极大圆与 c 不再相交.

1.2.5 定理 4: Moon in the Puddle (Pestov & Ionin, 1959)

设 $c: I \rightarrow \mathbb{R}^2$ 是一条简单正规回路, 若

$$\kappa(t) \leq 1, \quad \forall t \in I$$

则 c 所围区域一定包含一个单位圆 (unit disc).

其中, " $\kappa(t) \leq 1$ " 为曲线的局部性质, "一定包含一个单位圆" 为曲线的整体性质, 该定理将二者关联了起来.

假设曲线 c 上存在一点 $p = c(t_0)$, 使得该点处的密切圆 σ 位于曲线内部, σ 的半径 $r = \frac{1}{\kappa(t_0)} \geq 1$ 因此, 其一定包含一个单位圆.

1.2.6 定义 9: 环绕数 (Winding Number)

设 $c: [a, b] \rightarrow \mathbb{R}^2$ 为一个环路, P_0 为不在 c 上的一点, 将 c 沿逆时针绕过 P_0 圈数称为环绕数, 记作 $W(c, P_0)$. 显然有 $W(c, P_0) \in \mathbb{Z}$.

1.2.7 命题 4: 环绕数的性质

1. $W(c, P_0)$ 的值依赖于 P_0 的选取;
2. 若 P_1 与 P_2 能被一条不与 c 相交的曲线相连, 则有 $W(c, P_1) = W(c, P_2)$;
3. 若 c 的连续形变不经过 P_0 , 则 $W(c, P_0)$ 在该形变下保持不变, 即: 设 c_δ 为一族回路, 且有

$$c_\delta(t) \neq P_0, \quad \forall \delta, t$$

则 $W(c_\delta, P_0)$ 的取值与 δ 无关.

1.2.8 环绕数公式

$$W(c, O) = \frac{1}{2\pi} \int_a^b \frac{x(t)y'(t) - y(t)x'(t)}{x(t)^2 + y(t)^2} dt \quad (1.2.4)$$

将 P_0 平移到坐标原点 O , 替换到极坐标系下, 有 $c(t) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$, 回路 c 绕 O 的环绕数为 $W(c, O) = \frac{\theta(b) - \theta(a)}{2\pi}$,

$$\begin{aligned} \dot{c}(t) \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} &= \begin{pmatrix} \dot{r} \cos \theta - r \dot{\theta} \sin \theta \\ \dot{r} \sin \theta + r \dot{\theta} \cos \theta \end{pmatrix} = \dot{r} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} + r \dot{\theta} \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} = \frac{\dot{r}}{r} \begin{pmatrix} x \\ y \end{pmatrix} + \dot{\theta} \begin{pmatrix} -y \\ x \end{pmatrix} \\ \Rightarrow \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} (-y, x) &= \frac{\dot{r}}{r} \begin{pmatrix} x \\ y \end{pmatrix} (-y, x) + \dot{\theta} \begin{pmatrix} -y \\ x \end{pmatrix} (-y, x) \Rightarrow \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{x^2 + y^2} \\ \Rightarrow W(c, O) &= \frac{1}{2\pi} \int_a^b \frac{x\dot{y} - y\dot{x}}{x^2 + y^2} dt \end{aligned}$$

1.3 三维空间中曲线的性质

1.3.1 定义 10: $\mathbb{R}^3$ 中曲线的 Frenet-Serret 标架

设 c 为 $\mathbb{R}^3$ 中的参数化曲线, 其切向量为 $T(t)$, 曲率 $\kappa(t) = |\ddot{c}(t)|$. 假设 $\kappa > 0$, 则 $N(t) = \frac{\ddot{c}(t)}{|\ddot{c}(t)|}$, 且有副法向量 (Binormal Vector):

$$B(t) = T(t) \times N(t) \quad (1.3.1)$$

称 $\{T, N, B\}$ 为 $\mathbb{R}^3$ 中曲线的 Frenet-Serret 标架.

1.3.2 定义 11: 挠率 (Torsion)

设 c 为 $\mathbb{R}^3$ 中的弧长参数化曲线, N, B 分别为 c 的主法向量和副法向量, 称

$$\tau(s) = -\dot{B}(s) \cdot N(s) \quad (1.3.2)$$

为 c 在 s 处的挠率.

1.3.3 定理 5: Frenet-Serret 公式

设参数化曲线 $c: I \rightarrow \mathbb{R}^3$ 有 Frenet-Serret 标架 T, N, B , 则有

$$\frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = (\tau T + \kappa B) \times \begin{pmatrix} T \\ N \\ B \end{pmatrix} \quad (1.3.3)$$

$\dot{T} = \ddot{c} = \kappa N$, 假设存在 $a, b, c \in \mathbb{R}$ 使得 $\dot{N} = aT + bN + cB$,

$$\begin{cases} N \cdot T = 0 \xrightarrow{\frac{d}{dt}} \underbrace{\dot{N} \cdot T}_a + \underbrace{N \cdot \dot{T}}_{\kappa} = 0 & \implies a = -\kappa \\ N \cdot N = 1 \xrightarrow{\frac{d}{dt}} 2 \underbrace{\dot{N} \cdot N}_b = 0 & \implies b = 0 \\ N \cdot B = 0 \xrightarrow{\frac{d}{dt}} \underbrace{\dot{N} \cdot B}_c + N \cdot \dot{B} = 0 & \implies c = -N \cdot \dot{B} \end{cases}$$

令 $c = \dot{N} \cdot B = -N \cdot \dot{B} \equiv \tau$, 设 $\dot{B} = \tilde{a}T + \tilde{b}N + \tilde{c}B$,

$$\begin{cases} B \cdot B = 1 \xrightarrow{\frac{d}{dt}} 2 \underbrace{\dot{B} \cdot B}_{\tilde{c}} = 0 & \implies \tilde{c} = 0 \\ B \cdot T = 0 \xrightarrow{\frac{d}{dt}} \underbrace{\dot{B} \cdot T}_{\tilde{a}} + \underbrace{B \cdot \dot{T}}_{=B \cdot \kappa N=0} = 0 & \implies \tilde{a} = 0 \\ B \cdot N = 0 \xrightarrow{\frac{d}{dt}} \underbrace{\dot{B} \cdot N}_{\tilde{b}} + \underbrace{B \cdot \dot{N}}_{\tau} = 0 & \implies \tilde{b} = -\tau \end{cases}$$

$$\implies \begin{cases} \dot{T} = \kappa N \\ \dot{N} = -\kappa T + \tau B \\ \dot{B} = -\tau N \end{cases} \implies \frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

$$\begin{aligned} \frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} &= \begin{pmatrix} \kappa N \\ -\kappa T + \tau B \\ -\tau N \end{pmatrix} = \begin{pmatrix} \kappa B \times T \\ -\kappa N \times B + \tau T \times N \\ -\tau B \times T \end{pmatrix} \\ &= \begin{pmatrix} \tau T \times T + \kappa B \times T \\ \tau T \times N + \kappa B \times N \\ \tau T \times B + \kappa B \times B \end{pmatrix} = (\tau T + \kappa B) \times \begin{pmatrix} T \\ N \\ B \end{pmatrix} \end{aligned}$$

1.3.4 曲率与挠率的几何意义

曲率 κ 度量了标架绕 B 的转动; 而挠率 τ 度量了标架绕 T 的转动.

$$\Omega = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \quad (1.3.4)$$

显然 Ω 是一个斜对称矩阵 (skew-symmetric matrix):

$$\Omega^T = -\Omega \quad (1.3.5)$$

Ω 描述了标架如何变化

1.3.5 曲线参数方程在一点的标准展开

在 $t = 0$ 附近对一条弧长参数化曲线作 Taylor 展开:

$$c(t) = c(0) + t\dot{c}(0) + \frac{t^2}{2}\ddot{c}(0) + \frac{t^3}{6}\dddot{c}(0) + \cdots \quad (1.3.6)$$

$$\begin{aligned} \dot{c}(0) &= T(0) \\ \ddot{c}(0) &= \kappa(0)N(0) \\ \dddot{c}(0) &= \dot{\kappa}(0)N(0) - \kappa(0)^2T(0) + \kappa(0)\tau(0)B(0) \end{aligned} \quad (1.3.7)$$

$$c(t) \approx c(0) + tT(0) + \frac{t^2}{2}\kappa(0)N(0) + \frac{t^3}{6}\kappa(0)\tau(0)B(0) \quad (1.3.8)$$

1. $c(0) + tT(0)$ 表示曲线在切方向上的线性运动;
2. $c(0) + tT(0) + \frac{t^2}{2}\kappa(0)N(0)$ 表示曲线在 T - N 平面上的抛物运动 (取决于曲率 κ);
3. $\frac{t^3}{6}\kappa(0)\tau(0)B(0)$ 表示垂直于 T - N 平面的运动 (取决于挠率 τ).

1.3.6 命题 5: 空间曲线平面性的判定

设 $c: I \rightarrow \mathbb{R}^3$ 是一条单位速率曲线, 其曲率 $\kappa > 0$, 当且仅当挠率 $\tau = 0$ 时, c 位于一个平面内.

充分性: 若 c 位于一个平面内, 该平面必然是由 T, N 张成的, 因此 B 为一个方向固定的常向量, 于是

$$\dot{B} = -\tau N = 0 \xrightarrow{N \neq 0} \tau = 0$$

必要性: $\tau = 0 \implies \dot{B}(t) = 0 \implies B(t) \equiv B$, 令 $f(t) = (c(t) - c(0)) \cdot B$, 有

$$\frac{df}{dt} = \dot{c} \cdot B + c \cdot \dot{B} = T \cdot B = 0 \implies f(t) \equiv f(0) = 0$$

因此 c 位于与 B 垂直的平面上.

1.3.7 定理 6: 曲线论基本定理 (Frenet-Serret 定理)

若给定可微函数 $\begin{cases} \kappa: [0, T] \rightarrow (0, \infty) \\ \tau: [0, T] \rightarrow \mathbb{R} \end{cases}$, 点 $p_0 \in \mathbb{R}^3$, 与标准正交基 $T_0, N_0, B_0 \in \mathbb{R}^3$,

则存在唯一的单位速率曲线 $c: [0, T] \rightarrow \mathbb{R}^3$, 使得:

1. 其曲率为 $\kappa(t)$, 挠率为 $\tau(t)$;
2. 初始点为 $c(0) = p_0$;
3. 初始 F-S 标架为 $T(0) = T_0, N(0) = N_0, B(0) = B_0$.

该定理表明: 曲线的曲率和挠率完全决定了曲线的形状 (或称在刚体运动下唯一).

唯一性: 反证法. 假设存在两条满足条件的曲线 c 与 $\bar{c}$, 即

$$\begin{cases} \kappa_1(t) = \kappa_2(t) = \kappa(t), \tau_1(t) = \tau_2(t) = \tau(t) \\ c_1(0) = c_2(0) = p_0 \\ \{T(0), N(0), B(0)\} = \{\bar{T}(0), \bar{N}(0), \bar{B}(0)\} = \{T_0, N_0, B_0\} \end{cases}$$

$$\text{设 } f(t) = \frac{1}{2}|T(t) - \bar{T}(t)|^2 + \frac{1}{2}|N(t) - \bar{N}(t)|^2 + \frac{1}{2}|B(t) - \bar{B}(t)|^2$$

$$\dot{f}(t) = (\dot{T} - \dot{\bar{T}}) \cdot (T - \bar{T}) + (\dot{N} - \dot{\bar{N}}) \cdot (N - \bar{N}) + (\dot{B} - \dot{\bar{B}}) \cdot (B - \bar{B}) \stackrel{\text{F-S eqs.}}{=} 0$$

$$\stackrel{f(0)=0}{\implies} f(t) \equiv 0 \implies \begin{cases} T(t) = \bar{T}(t) \\ N(t) = \bar{N}(t) \\ B(t) = \bar{B}(t) \end{cases} \implies \dot{c}(t) = \dot{\bar{c}}(t) \stackrel{c(0)=\bar{c}(0)}{\implies} c(t) = \bar{c}(t)$$

存在性: 给定常微分方程组

$$\frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \quad (1.3.9)$$

初始条件为 $T(0) = T_0, N(0) = N_0, B(0) = B_0$. 由于 $\kappa(t), \tau(t)$ 为可微函数, 满足连续性与 Lipschitz 条件, 根据一阶常微分方程组解的存在唯一性定理 (Picard-Lindelöf 定理), 存在满足初始条件的唯一的解 $\{T(t), N(t), B(t)\}$. 于是可定义曲线:

$$c(t) = p_0 + \int_0^t T(\tau) d\tau$$

注记

通常, 确定一条曲线需要三个函数, 即 $c(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$. 在该定理中, 三个函数分别为:

κ, τ 与 $v \equiv 1$.

2 曲面论

概念	曲线	曲面
定义域	区间 $I \subseteq \mathbb{R}$	开子集 $D \subseteq \mathbb{R}^2$
参数化	$c : I \rightarrow \mathbb{R}^3$	$\Phi : D \rightarrow \mathbb{R}^3$
正规性	$\dot{c}(t) \neq 0$	$T_p\Phi$ 是单射
无自交	c 是单射	Φ 是单射

2.1 曲面的定义与构造

2.1.1 定义 1: 开球 (Open Ball), 开圆盘 (Open Disc)

设 $p = (p_1, p_2) \in \mathbb{R}^2$, 则称

$$B_\varepsilon(p) = \{x = (x_1, x_2) \in \mathbb{R}^2 \mid |x - p|^2 = (x_1 - p_1)^2 + (x_2 - p_2)^2 < \varepsilon^2\} \quad (2.1.1)$$

为 $\mathbb{R}^2$ 中的开球或开圆盘.

开球的概念可推广至 $\mathbb{R}^n$, 开圆盘为开球在 $\mathbb{R}^2$ 的特例.

2.1.2 定义 2: $\mathbb{R}^2$ 中的开集 (Open Subset)

设一个集合 $D \in \mathbb{R}^2$, 若对于每一点 $p \in D$, 都存在 $\varepsilon > 0$, 使得 D 包含开球 $B_\varepsilon(p)$, 则称 D 为 $\mathbb{R}^2$ 中的一个开集.

2.1.3 定义: 切映射 (Tangent Map)

设开集 $D \in \mathbb{R}^2$, 映射 $F : D \rightarrow \mathbb{R}^m$, 若对任意 $p \in D$ 与 $v \in \mathbb{R}^2$, 极限

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} F(p + \varepsilon v) = \lim_{\varepsilon \rightarrow 0} \frac{F(p + \varepsilon v) - F(p)}{\varepsilon} \quad (2.1.2)$$

存在, 则称 F 在 p 处沿 v 方向有方向导数, 记为 $T_p F(v)$. 由此得到的映射 $T_p F : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ 称为 F 在 p 处的切映射.

2.1.4 定义 3: 高维映射的可微性 (Differentiability) 与光滑性 (Smoothness)

设一个连续映射 $\Phi : D \rightarrow \mathbb{R}^3$ ($D \in \mathbb{R}^2$), 若对于每一点 $p = (p_1, p_2) \in D$, 都存在一个开球 $B_r(p) \subseteq D$, 使得映射

$$\begin{aligned} c_x : (-r, +r) &\rightarrow \mathbb{R}^3, c_x(t) = \Phi(p_1 + t, p_2) \\ c_y : (-r, +r) &\rightarrow \mathbb{R}^3, c_y(t) = \Phi(p_1, p_2 + t) \end{aligned} \quad (2.1.3)$$

均可微, 则称 Φ 是可微的.

2.1.5 定义 4: 高维映射的偏导数 (Partial Derivative)

映射 Φ 在 $p \in D$ 处的偏导数定义为:

$$\begin{aligned}\frac{\partial \Phi}{\partial x} &= \left. \frac{d}{dt} \right|_{t=0} c_x(t) \\ \frac{\partial \Phi}{\partial y} &= \left. \frac{d}{dt} \right|_{t=0} c_y(t)\end{aligned}\tag{2.1.4}$$

若 $\Phi(x, y) = \begin{pmatrix} \Phi_1(x, y) \\ \Phi_2(x, y) \\ \Phi_3(x, y) \end{pmatrix}$, 则 $\frac{\partial \Phi}{\partial x} = \begin{pmatrix} \frac{\partial \Phi_1}{\partial x}(x, y) \\ \frac{\partial \Phi_2}{\partial x}(x, y) \\ \frac{\partial \Phi_3}{\partial x}(x, y) \end{pmatrix}$. 于是, 映射的偏导数可构成新的映射:

$$\begin{aligned}\frac{\partial \Phi}{\partial x} : D \rightarrow \mathbb{R}^3 \quad (D \in \mathbb{R}^2) \\ (p_1, p_2) \mapsto \frac{\partial \Phi}{\partial x}(p_1, p_2)\end{aligned}\tag{2.1.5}$$

2.1.6 定义 6: 高维映射的 Jacobian 矩阵

Φ 在 p 处的 Jacobian 矩阵为:

$$\mathbf{J}_\Phi(p) = \left(\frac{\partial \Phi}{\partial x}, \frac{\partial \Phi}{\partial y} \right) = \begin{pmatrix} \frac{\partial \Phi_1}{\partial x} & \frac{\partial \Phi_1}{\partial y} \\ \frac{\partial \Phi_2}{\partial x} & \frac{\partial \Phi_2}{\partial y} \\ \frac{\partial \Phi_3}{\partial x} & \frac{\partial \Phi_3}{\partial y} \end{pmatrix}\tag{2.1.6}$$

$\Phi : D \rightarrow \mathbb{R}^3 \ (D \in \mathbb{R}^2) \implies$ Jacobian 矩阵是一个 3×2 矩阵.

2.1.7 定义 7: 高维映射的正规性 (Regularity)

Φ 是正规的, 或者称非退化的 (non-degenerate), 当且仅当:

1. $\forall p \in D$, 切映射 $T_p \Phi$ 的像均是 2 维的 (正规的参数化曲面 Φ 每一点都有唯一的切平面);
2. 偏导数 $\frac{\partial \Phi}{\partial x}(p)$ 与 $\frac{\partial \Phi}{\partial y}(p)$ 线性无关, 即二者非零且不平行;
3. $\forall p \in D, \text{rank } \mathbf{J}_\Phi(p) = 2$;

以上均为正规性的等价表述.

2.1.8 定义 8: 参数化曲面 (Parameterized Surface)

称一个定义在开集 $D \subseteq \mathbb{R}^2$ 上的光滑正规单射 $\Phi: D \rightarrow \mathbb{R}^3$ 为**参数化曲面**, 其像集

$$S = \Phi(D) = \{\Phi(x, y) \mid (x, y) \in D\} \quad (2.1.7)$$

也称为**参数化曲面**.

2.1.9 定义: 坐标卡 (Coordinate Chart)

给定 $\mathbb{R}^3$ 的一个子集 S , 围绕点 $p \in S$ 的一个**坐标卡** Φ 包含:

1. 一个开集 $D \subseteq \mathbb{R}^2$;
2. 一个光滑正规单射 $\Phi: D \rightarrow \mathbb{R}^3$;
3. 一个包含 p 的开集 $\mathcal{O} \subseteq \mathbb{R}^3$ (p 的邻域),

满足 $\Phi(D) = \{\Phi(p) \mid p \in D\} = S \cap \mathcal{O}$, 且逆映射 $\Phi^{-1}: S \cap \mathcal{O} \rightarrow D$ 连续. 若 $0 \in D$ 且 $\Phi(0) = p$, 则称该坐标卡以 p 为中心.

坐标卡 Φ 即对 S 的一部分 $\Phi(D)$ 的一个参数化.

2.1.10 定义: 曲面 (Surface)

设 S 为 $\mathbb{R}^3$ 的一个子集, 若每一点 $p \in S$ 都有以 p 为中心的坐标卡, 则称 S 是一个**曲面**.

即, 曲面不仅是 $\mathbb{R}^3$ 的子集, 还需配备一组能覆盖整个曲面的坐标卡.

曲面为三维 Euclidean 空间 $\mathbb{R}^3$ 中具有局部平坦性的二维子集, 即曲面的每一点附近, 都能找到一个足够小的邻域, 使其可平滑展开为平面的一部分, 且无折叠, 无断裂, 无退化.

2.1.11 定义: 坐标变换 (Coordinate Transition)

设 $\Phi: D \rightarrow S \cap \mathcal{O}$ 与 $\Psi: E \rightarrow S \cap \mathcal{U}$ 是曲面 S 上两个有重叠的坐标卡 ($S \cap \mathcal{O} \cap \mathcal{U} \neq \emptyset$), 则映射

$$\Psi^{-1} \circ \Phi: \Phi^{-1}(S \cap \mathcal{O} \cap \mathcal{U}) \rightarrow \Psi^{-1}(S \cap \mathcal{O} \cap \mathcal{U}) \quad (2.1.8)$$

称为**坐标变换**.

2.1.12 命题 1: 函数图像构造法 (Graph)

设 $D \subseteq \mathbb{R}^2$ 为一个开集, $f: D \rightarrow \mathbb{R}$ 为一个光滑函数, 则 f 的**图像**:

$$S = \text{graph of } f = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, z = f(x, y)\} \quad (2.1.9)$$

为一个参数化曲面, 且存在全局坐标卡

$$\Phi(x, y) = (x, y, f(x, y)) \quad (x, y) \in D \quad (2.1.10)$$

2.1.13 命题 2: 水平集构造法 (Level Set)

设 $g: \mathbb{R}^3 \rightarrow \mathbb{R}$ 是一个光滑函数, 对于任意 $c \in \mathbb{R}$, 若水平集

$$S = g^{-1}(c) = \{(x, y, z) \in \mathbb{R}^3 \mid g(x, y, z) = c\} \quad (2.1.11)$$

满足对所有 $p \in S$, 都有

$$\nabla g(p) = \left(\frac{\partial g}{\partial x}(p), \frac{\partial g}{\partial y}(p), \frac{\partial g}{\partial z}(p) \right) \neq 0 \quad (2.1.12)$$

则 S 是一张曲面.

2.2 切空间与向量场

2.2.1 定义 1: 表面上的光滑函数 (Smooth Function on Surface)

设 $f: S \rightarrow \mathbb{R}^k$ 是一个定义在曲面 S 上的光滑函数, 若对于每一个坐标卡 $\Phi: D \rightarrow S$, 复合映射 $f \circ \Phi: D \rightarrow \mathbb{R}^k$ 是通常意义上的光滑函数, 则称 f 是 S 上的光滑函数. $f \circ \Phi$ 为 f 的卡表示 (chart representation), 记作 f_Φ .

2.2.2 定义 2: 表面上的光滑曲线 (Smooth Curves on Surface)

若光滑的参数化曲线 $c: I \rightarrow \mathbb{R}^3$ 满足: $c(t) \in S, \forall t \in I$, 则称其为曲面 S 上的光滑曲线.

也可定义为映射 $c: I \rightarrow S$, 且对于 S 上每个坐标卡 Φ , $c_\Phi = \Phi^{-1} \circ c: I_\Phi \rightarrow \mathbb{R}^3$ 光滑, 其中 $I_\Phi = c^{-1}(\Phi(D))$ 为 I 映射到卡 $\Phi(D)$ 的部分.

2.2.3 定义 3: 切空间 (Tangent Space)

给定曲面 $S \in \mathbb{R}^3$ 与一点 $p \in S$, 若存在一条光滑曲线 $c: I \rightarrow S$, 满足 $c(0) = p$ 且 $\dot{c}(0) = X$, 则称向量 $X \in \mathbb{R}^3$ 为 S 在 p 处的切向量. S 在 p 处所有这样的切向量构成的集合, 称为 S 在 p 处的切空间 $T_p S$.

直观上, 切空间 $T_p S$ 是在 p 处与 S 恰好相切的平面.

2.2.4 引理

对曲面 S 上的任意一点, 与任意以 p 为中心的坐标卡 $\Phi: D \rightarrow S$, 切空间 $T_p S$ 由切向量 $\partial_x^\Phi(p)$ 和 $\partial_y^\Phi(p)$ 张成. 即, 对于任意 $X \in \mathbb{R}^3$, 存在唯一的 $X_1, X_2 \in \mathbb{R}$, 使得

$$X = X_1 \partial_x^\Phi(p) + X_2 \partial_y^\Phi(p) \quad (2.2.1)$$

特别地, $T_p S$ 是一个 2 维向量空间.

2.2.5 定义: 向量场 (Vector Field)

$X: S \rightarrow \mathbb{R}^3$ 使得 $X(m) \in T_m S, \forall m \in S$

2.3 微分形式

Question: how do we measure the area of a surface? Or the length of a curve on a surface?

$$L(c) = \int_0^1 |\dot{c}(\tau)| d\tau$$

Similarly, we could "define" the area of S : $A(S) = \dots$

Need to measure the area of the parallelogram spa by two tangent vector. Another instance, where we want to measure the "length" of a vector.

Differential forms are gadgets that eat tangent vectors and produce a number.

2.3.1 定义: 1-形式

A differential 1-form on S is a linear map $\alpha_m: T_m S \rightarrow \mathbb{R}$ ($\forall m \in S$). We write α for the whole family, and $\alpha_m: T_m S \rightarrow \mathbb{R}$.

Visualization

$\alpha(X_m) = \star$ of intersections of the planes α_m with the vector X_m .

2.3.2 Differential forms on $D \subseteq \mathbb{R}^2$

Every tangent vector of $\mathbb{R}^2$ is of the form $X_{(x,y)} = X_1(x,y)\partial_x(x,y) + X_2(x,y)\partial_y(x,y)$

Let $\alpha_{(x,y)}: T_{(x,y)} D \rightarrow \mathbb{R}$ be a 1-form, then

$$\alpha_{(x,y)}(X_{(x,y)}) = \alpha_{(x,y)}(X_1(x,y)\partial_x + X_2(x,y)\partial_y) = X_1(x,y)\alpha_{(x,y)}(\partial_x) + X_2(x,y)\alpha_{(x,y)}(\partial_y)$$

$\implies \alpha_{(x,y)}$ is uniquely defined by two numbers $\alpha_1(x,y) = \alpha_{(x,y)}(\partial_x)$ and $\alpha_2(x,y) = \alpha_{(x,y)}(\partial_y)$

$$\implies \alpha_{(x,y)}(X) = X_1(x,y)\alpha_1(x,y) + X_2(x,y)\alpha_2(x,y)$$

$\implies \alpha_1, \alpha_2$ are called the **components** of α .

Different way to view α_1, α_2 : which is similar to $X = x_1\partial_x + X_2\partial_y$

Define 1-form $(dx)_{(x,y)}: T_{(x,y)}D \rightarrow \mathbb{R}$, by $(dx)_{(x,y)}(\partial_x) = 1$, $(dx)_{(x,y)}(\partial_y) = 0$. Similarly, the 1-form dy is defined by $dy(\partial_x) = 0$, $dy(\partial_y) = 1$.

$$\implies dx(X) = X_1, dy(X) = X_2$$

$$\alpha_{(x,y)}(X_{(x,y)}) = X_1\alpha_1 + X_2\alpha_2 = (dx)_{(x,y)}(X_{(x,y)})\alpha_1(x,y) + (dy)_{(x,y)}(X_{(x,y)})\alpha_2(x,y)$$

$$\implies \alpha_{(x,y)} = \alpha_1(x,y)(dx)_{(x,y)} + \alpha_2(x,y)(dy)_{(x,y)}$$

Every differential form α on D is of the form $\alpha = \alpha_1 dx + \alpha_2 dy$, for (smooth) functions α_1, α_2 on D .

Remark: 1-forms are elements in the dual space of $T_{(x,y)}D$ and dx & dy is the basis dual to ∂_x & ∂_y .

If X is a vector field on S and α is 1-form on S , then $\forall m \in S$, we can plug in the vector $X_m \in T_m S$ into $\alpha_m: T_m S \rightarrow \mathbb{R}$ to get $\alpha_m(X_m) \in \mathbb{R}$. That defines us a function $\alpha(X): S \rightarrow \mathbb{R}$

$$(\alpha(X))(m) = \alpha_m(X_m)$$

2.3.3 Chart Representation of a 1-form

α is a 1-form on S and $\Phi: D \rightarrow S$ is a chart on S , $T_{(x,y)}\Phi: \mathbb{R}^2 \rightarrow T_{\Phi(x,y)}S$ and $\alpha_{\Phi(x,y)}: T_{\Phi(x,y)}S \rightarrow \mathbb{R}$. the composition on $\alpha_{\Phi(x,y)}^{\Phi} = \alpha_{\Phi(x,y)} \circ T_{(x,y)}\Phi: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a linear map, i.e. α^{Φ} is a 1-form on D .

The components of α^{Φ} are

$$\alpha_{(x,y)}^{\Phi}(\partial_x(x,y)) = \alpha_{\Phi(x,y)}(\underbrace{T_{(x,y)}\Phi(\partial_x(x,y))}_{\partial_x^{\Phi}(\Phi(x,y))}) = (\alpha(\partial_x^{\Phi}))(\Phi(x,y))$$

$\implies$ The component $\alpha^{\Phi}(\partial_x)$ is the chart representation of $\alpha(\partial_x^{\Phi})$

Differential 1-forms can be integrated along 1-dim objects on the surface.

2.3.4 Integral

If α is 1-form on S and $c: [a,b] \rightarrow S$ is a smooth curve on S , then the integral of α along c is defined by

$$\int_c \alpha := \int_a^b \alpha_{c(t)}(\dot{c}(t)) dt$$

$$\alpha_{c(t)}: T_{c(t)}S \rightarrow \mathbb{R}$$

2.3.5 Last time

Exterior diff

$$f : S \rightarrow \mathbb{R}, (df)_m(X) = \left. \frac{d}{dt} \right|_0 f(c(t)), \dot{c}(0) = X$$

Integral of 1-form α along $c : [0, T] \rightarrow S$

$$\int_c \alpha = \int_0^T \alpha_{c(t)}(\dot{c}(t)) dt$$

2.3.6 Prop: Fundamental Theorem

If $f : S \rightarrow \mathbb{R}$ is a smooth function and $c : [0, T] \rightarrow S$ a smooth curve, then

$$\int_c df = f(c(T)) - f(c(0))$$

Proof $\int_c df = \int_0^T (df)_{c(t)}(\dot{c}(t)) dt = \int_0^T \left. \frac{d}{dt} \right|_t f(c(t)) dt$ (Fundamental Theorem in Calculus)

2.3.7 Prop:

$\Phi : [a', b'] \rightarrow [a, b]$ smooth s.t. $\Phi'(s) \neq 0, \forall s \in [a', b']$, then

$$\int_{c \circ \Phi} \alpha = \begin{cases} \int_c \alpha & \text{if } \Phi'(s) > 0 \\ - \int_c \alpha & \text{if } \Phi'(s) < 0 \end{cases}$$

Example Let $c : [a, b] \rightarrow \mathbb{R}^2 - \{0\}$ be closed curve write as $c(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$. Define $\alpha = \frac{1}{2\pi} \frac{-ydx + xdy}{x^2 + y^2}$ (i-form $\in \mathbb{R}^2 - \{0\}$)

$$\int_c \alpha = \frac{1}{2\pi} \int_0^T \frac{-y(t)dx(\dot{c}(t)) + x(t)dy(\dot{c}(t))}{x(t)^2 + y(t)^2} dt = W(c, 0) \in \mathbb{Z}$$

2.4 Differential 2-form

1-form eats 1 vector, gives a number

2-form eats 2 vector

2.4.1 Def: Differential 2-form & Differential k -form

A differential 2-form β on surface S is, for each point $m \in S$, a bilinear map

$$\beta_m : T_m S \times T_m S \rightarrow \mathbb{R}$$

s.t. $\beta_m(v, w) = -\beta_m(w, v) \forall v, w \in T_m S$

A differential k -form is a map

$$\gamma_m : \underbrace{T_m S \times \cdots \times T_m S}_{k\text{-times}} \rightarrow \mathbb{R}$$

1. multilinear

2. skew-symmetric (like the wave-function of fermions)

$$\begin{aligned} \beta_m(av + bw, cv + dw) &= \underbrace{ac\beta_m(v, v)}_{=0} + ad\beta_m(v, w) + bc\beta_m(w, v) + \underbrace{bd\beta_m(w, w)}_{=0} = (ad - \\ bc)\beta_m(v, w) &= \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \beta_m(v, w) \end{aligned}$$

2.4.2 Constructions of 2-forms

α_1 and α_2 are 1-form, then $\alpha_1 \wedge \alpha_2$ is defined by

$$(\alpha_1 \wedge \alpha_2)_m(v, w) = (\alpha_1)_m(v)(\alpha_2)_m(w) - (\alpha_1)_m(w)(\alpha_2)_m(v), v, w \in T_m S$$

Note $\alpha_1 \wedge \alpha_2 = -\alpha_2 \wedge \alpha_1$

Similarly α a 1-form and β a 2-form then $\alpha \wedge \beta$ is 3-form def by

$$(\alpha \wedge \beta)_m(v, w, u) = \alpha_m(v)\beta_m(w, u) - \alpha_m(w)\beta_m(v, u) - \alpha_m(u)\beta_m(v, w)$$

Example on $\mathbb{R}^3$, $\alpha = a_1 dx + a_2 dy + a_3 dz$, $\beta = b_1 dx + b_2 dy + b_3 dz$ then

$$\alpha \wedge \beta = (a_2 b_3 - a_3 b_2) dx \wedge dz + (a_3 b_1 - b_3 a_1) dz \wedge dx + (a_1 b_2 - a_2 b_1) dx \wedge dy$$

the coefficients are exactly the coeff. of $\vec{a} \times \vec{b}$

If $\gamma = c_1 dx + c_2 dy + c_3 dz$, then $\alpha \wedge \beta \wedge \gamma = (\text{coff of } \vec{c} \times (\vec{a} \times \vec{b})) dx \wedge dy \wedge dz$

Exterior derivative of a 1-form

Notivation If $f : S \rightarrow \mathbb{R}$, Then $df(X)$ encodes how f changes in the di. X

If α is a 1-form, then $d\alpha(X, Y)$ is a 2-form that encodes the change of α in a given direction.

A natural guess is X as the direction of the change of $f(p) = \alpha_p(Y)$

Have to satisfy

$$(d\alpha)_m(X, Y) = T_m(\alpha(Y))(X) - T_m(\alpha(X))(Y), \quad X, Y \in T_m S$$

Problem $f(p) = \alpha_p(Y)$ doesn't make sense as $Y \in T_m S$

$\implies$ Need a vector field Y instead of only a single fixed vector

$\implies$ Most convenient choice: choose vector field associated with the charts

2.4.3 Def: Exterior Derivative of 1-form

α is a 1-form on S . The exterior derivative $d\alpha$ is a 2-form on S defined by

$$d\alpha \left(\partial_x^\Phi(m), \partial_y^\Phi(m) \right) = T_m \left(\alpha(\partial_y^\Phi) \right) (\partial_x^\Phi(m)) - T_m \left(\alpha(\partial_x^\Phi) \right) (\partial_y^\Phi(m))$$

for a chart $\Phi : D \rightarrow S$, $D \subseteq \mathbb{R}^2$

Remark

1. Enough to specify $d\alpha$ on ∂_x^Φ and ∂_y^Φ by bilinearity and skew-symmetric and since $\partial_x^\Phi(m)$ and $\partial_y^\Phi(m)$
2. Def doesn't depend on Φ

Let's analyze $T_m f \left(\partial_x^\Phi(m) \right)$ for $f = \alpha \left(\partial_y^\Phi \right)$

Recall $T_m \left(\partial_x^\Phi \right) = \frac{\partial f^\Phi}{\partial x}$ for $f^\Phi = f \circ \Phi$ being the chart representation of f

What the chart representation of $f = \alpha \left(\partial_y^\Phi \right)$?

It is the y -component of the chart representation α^Φ of α

$$\alpha^\Phi = \alpha_1 dx + \alpha_2 dy$$

$(\alpha^\Phi: 1\text{-form} \in D, \alpha_2 = f \circ \Phi)$

$$f^\Phi = \alpha^\Phi(\partial_y)$$

In summary $(d\alpha)_m \left(\partial_x^\Phi(m), \partial_y^\Phi(m) \right) = \frac{\partial}{\partial x} \left(\alpha^\Phi(\partial_y) \right) - \frac{\partial}{\partial y} \left(\alpha^\Phi(\partial_x) \right)$ at x, y s.t. $\Phi(x, y) = m$

two ways to calculate $d\alpha$

1. on S : calculate $\partial_x^\Phi, \partial_y^\Phi$, then plug in α , then take derivative
2. on D : calculate α^Φ and then are $\dots$

If $S = D$ an open subset of $\mathbb{R}^2$, α is 1-form, $\alpha = \alpha_1 dx + \alpha_2 dy$ for $\alpha_1, \alpha_2 : S \rightarrow \mathbb{R}$

then ... claim

$$d\alpha = d\alpha_1 \wedge dx + d\alpha_2 \wedge dy$$

Evaluate side on ∂_x, ∂_y

$$(d\alpha_1 \wedge dx)(\partial_x, \partial_y) = d\alpha_1(\partial_x)dx(\partial_y) - d\alpha_1(\partial_y)dx(\partial_x) = -(d\alpha_1)(\partial_y) = -\frac{\partial\alpha_1}{\partial y}$$

Properties

Leibniz: $f : S \rightarrow \mathbb{R}$, α 1-form on S

$$d(f\alpha) = df \wedge \alpha + f d\alpha$$

$$d(df) = 0$$

Integravation on S Recall: 1-form α , $\int_c^T \alpha = \int_0^T \alpha(\dot{c}(t))dt$

A regular region in S is a smooth map $R : [a, b] \times [c, d] \rightarrow S$

2.4.4 Def: Integral

For a 2-form β on S , the integral of β over a rectangular region R is def by

$$\int_R \beta = \int_a^b \int_c^d \beta_{R(x,y)} \left(\frac{\partial R}{\partial x}(x, y), \frac{\partial R}{\partial y}(x, y) \right) dy dx$$

Hence $\frac{\partial R}{\partial x}$ is viewed on the tangent vector associated with the carve $t \mapsto R(x+t, y)$ in S

Example $R : [0, 1] \times [0, 2] \rightarrow \mathbb{R}^2$, $R(x, y) = (x, y)$

$$\frac{dR}{dx} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \partial_x$$

$$\int_R dx \wedge dy = \int_0^1 \int_0^2 (x dx \wedge dy)_{(x,y)} (\partial_x, \partial_y) dy dx = \int_0^1 \int_0^2 x dy dx = 1$$

2.4.5 Fundamental Theorem in Calculus

$$\int_c df = f(c(t)) - f(c(0))$$

Theorem (Stokes) $U \subseteq \mathbb{R}^2$ open and α a 1-form on U , Let $R : [a, b] \times [c, d] \rightarrow U$ be a rectangular region in U and let ∂R be the boundary curve, s.t. R lies to the left of ∂R

If R is injective and its Jacobian has positive det, then

$$\int_R d\alpha = \int_{\partial R} \alpha$$

Proof $\alpha = \alpha_1 dx + \alpha_2 dy$ for some $\alpha_1, \alpha_2 : U \rightarrow \mathbb{R}$ then

$$d\alpha = d\alpha_1 \wedge dx + d\alpha_2 \wedge dy = \left(\frac{\partial \alpha_2}{\partial x} - \frac{\partial \alpha_1}{\partial y} \right) dx \wedge dy$$

$$d\alpha_1 = \frac{\partial \alpha_1}{\partial x} dx + \frac{\partial \alpha_1}{\partial y} dy$$

$$\int_R d\alpha = \int_a^b \int_c^d (d\alpha)_{R(s,t)} \left(\frac{\partial R}{\partial s}, \frac{\partial R}{\partial t} \right) dt ds = \dots = \int_a^b \int_c^d \left(\frac{\partial \alpha_2}{\partial x} - \frac{\partial \alpha_1}{\partial y} \right) \cdot \det(\mathbf{J}) dt ds$$

What about integration over more general region².

Idea: patch up yow surface by rectangular regions and then sum their contributions.

2.4.6 Def: Compact

A surface is called compact is it can be covered by a finite number of rectangular regions.

Each of the rectangular region R has a which we can think of in two ways

1. the normal vector field $\frac{\partial R}{\partial x} \times \frac{\partial R}{\partial y}$ pointing outside
2. the orientation of the boundary area so that the inside lies left to the curve

2.5

2.5.1 Last time

For a surface $S \subseteq \mathbb{R}^3$ on a orientation is a

1. differentiable field of vectors in $\mathbb{R}^3$ $n : S \rightarrow \mathbb{R}^3$ which is nowhere vanishing and normal in the space

$$n(p) \cdot v = 0 \quad \forall p \in S, v \in T_p S$$

we call n the normal field

2. A of S into rectangular regions each is oriental in such a way that the boundary between two adjacent rectangles have the opposite orientation

Usually the normal field is $|n(p)| = 1$ in this case, we call n the unit normal field and the map

$$n : S \rightarrow S^2$$

is called the Gauss map

Not every surface has a orientation (e.g. the)

2.5.2 Def

If β is a 2-form on S , and R_i is a finite decomposition of S in oriented rectangular region, then

$$\int_S \beta = \sum_i \int_{R_i} \beta$$

Does NOT depend on the chosen decomposition

2.5.3 Theorem: Stokes

For any 1-form on a compact oriented surface S with boundary curve ∂S one has

$$\int_S d\alpha = \int_{\partial S} \alpha \quad (2.5.1)$$

Proof Apply the Stokes theorem to each R_i and observe that boundary contributions of the curves boundary areas cancel out, because they have the opposite orientation

Ex: Faraday's eq. can be written in the form

$$dE + \frac{\partial B}{\partial t} = 0$$

E electric field is a 1-form; B magnetic field is a 2-form

$$0 = \int_S dE + \int_S \frac{\partial B}{\partial t} = \int_{\partial S} E + \frac{d}{dt} \int_{\partial S} B$$

time varying magnetic field induces an electric field in the wire

$$\int_{\partial \Omega} \omega = \int_{\Omega} d\omega \quad (2.5.2)$$

Area element Look at chart first $\Phi : D \rightarrow S$ ($D \in \mathbb{R}^2$) What is the area of $\Phi(D)$?
On D we know how to calculate the area. Decompose D into rectangles and then sum them.
But under Φ each of the rectangles gets deformed.

The line segment from $\Phi(x, y)$ to $\Phi(x + v_1, y)$ is linearly by $T_{(x,y)}(v_1, D)$ and similarly $\Phi(x, y) \rightarrow \Phi(x, y + v_2)$ is approximated by $T_{(x,y)}\Phi(0, v_2)$.

So area of $\Phi(R) \approx \left| \left[T_{(x,y)}\Phi(v_1, 0) \right] \times \left[T_{(x,y)}\Phi(0, v_2) \right] \right|$ the area of the parallelogram spanned by $T\Phi(v_1, 0)$ and $T\Phi(0, v_2)$ so if we have a 2-form dA on S .

satisfying $(dA)_p(u_1, u_2) = |u_1 \times u_2|$ for $p \in S$, $u_1, u_2 \in T_p S$ (NOT linear).

Then area of $\Phi(R) \simeq (dA)_{\Phi(p)}(T_{(x,y)}\Phi(v_1, 0), T_{(x,y)}\Phi(0, v_2)) = (dA)_{(x,y)}^\Phi((v_1, 0), (0, v_2))$
and hence area of $\Phi(D) = \sum_R \text{area of } \Phi(R) = \int (dA)^\Phi = \int dA$.

2.5.4 Def

The area element on the surface S oriented by a unit normal field n is the 2-form dA def by $(dA)_p(u_1, u_2) = n_p \cdot (u_1 \times u_2)$, $p \in S, u_1, u_2 \in T_p S$.

the total area of a compact S is def by $\int_S dA$.

If we think of the orientation given by a decomposition in R_i then $dA(u_1, u_2) = \pm |u_1 \times u_2|$ with sign chosen in such a way that $\int_{R_i} dA$ is positive.

Ex. On the plane, $dA = dx \wedge dy$; on the sphere $dA(u_1, u_2) = p \cdot (u_1 \times u_2)$, $\int_S dA = 4\pi$

Remark the notation dA is misleading, since there is no 1-form A , since otherwise by Stokes $\int_S dA = \int_{\partial S} A = 0$ if S has no boundary.

2.6 Curvature of Surface

as for curves, we want to use frames to define curvature

Recall for curves, we need O, N, B , at each point of curve (T, N, B) , which is adapted to the curve ($\dot{c}(t) \sim T$). The rate of change of T, N, B told us the curvature and torsion (F-S eq.)

Want to do something similar for surfaces.

2.6.1 Def

A moving frame on an open subset $U \subseteq \mathbb{R}^3$ of vector fields $e_1, e_2, e_3 : U \rightarrow \mathbb{R}^3$, such that for each point $x \in U$, the vector $\{e_1(x), e_2(x), e_3(x)\}$ are a basis of $\mathbb{R}^3$. They are called ortho frame if $e_1(x), e_2(x), e_3(x)$ are TNB

F-S change of T, N, B are important. We define de_1 by apply d in every component

$$e_1(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \\ f_3(x) \end{pmatrix} \text{ for smooth functions } f_1, f_2, f_3 : U \rightarrow \mathbb{R}^3 \text{ then } de_1 = \begin{pmatrix} df_1 \\ df_2 \\ df_3 \end{pmatrix}$$

so we view de_1 as a linear map $\mathbb{R}^3 \rightarrow \mathbb{R}^3$. In particular, $v \in \mathbb{R}^3$, we have $(de_1)_x(v) \in \mathbb{R}^3$

So we can expand that vector with respect to the basis $e_1(x), e_2(x), e_3(x)$

$$de_1(v) = \omega_1^i(v) e_i$$

Left-hand side is linear in v , $\omega^1, \omega^2, \omega^3$ are 1-form on U , e_1, e_2, e_3 are orthonal frame

$$\omega_1^j(v) = de_1(v) \cdot e_j(x)$$

so ω_1^j tells you how e_1 is changing e_j -direction $\omega_1^j(v)$ measures how much e_1 rotates towards e_j if you move from x into the v -direction

More general

$$de_i(v) = \sum_j \omega_1^j(v) e_j$$

for ONB $\omega_1^j(v) = de_i(v) \cdot e_j$

In short, $de_1 = \sum_j \omega_1^j e_j$

so for all i and j , we get a 1-form ω_i^j on U . These are called the connection 1-form and is called the conn

so if e_i is ONB $e_i \cdot e_j$ is co function on U either 1 or 0

$$(de_i) \cdot e_j + e_i \cdot (de_j) = 0 \quad \omega_i^j = -\omega_j^i$$

so we have a skew-symmetric matrix $(\omega_i^j) = \begin{pmatrix} 0 & \omega_1^2 & \omega_1^3 \\ -\omega_1^2 & 0 & \omega_2^3 \\ -\omega_1^3 & -\omega_2^3 & 0 \end{pmatrix}$

Prop Let c be a unit-speed curve with $\kappa > 0$. Suppose e_1, e_2, e_3 is an orthonormal frame on $\mathbb{R}^3$ s.t. the of the fields to c give the Frenet frame T, N, B of c , then $(\omega_1^2)_{c(t)}(\dot{c}(t)) = \kappa$, $(\omega_1^3)_{c(t)}(\dot{c}(t)) = 0$, $(\omega_2^3)_{c(t)}(\dot{c}(t)) = \tau$

1. We recover curvature and torsion of a curve in frames of ω_i^j
2. We gained a lot flexibility, we can adopt the frame also to other geometric object (surface)

2.6.2 Dual f

the dual frame consists of 1-form ω^k defined by $\omega^k(e_i) = \begin{cases} 1 & \text{if } k = i \\ 0 & \text{otherwise} \end{cases}$

ω_i^k is the dual basis of e_i

Observation: for all $x \in U$ the map linear $v \mapsto e_k \omega^k \in \mathbb{R}^2$ is the identity map

Hence, it is independent of x , $o = d\left(\sum_k e_k \omega^k\right) = \sum_k de_k \wedge \omega^k + e_k d\omega^k = \sum_k \sum_i e_i (\omega_k^i \wedge \omega^k) + \sum_k e_k d\omega^k = \sum_k e_k (\sum_i \omega_k^i \wedge \omega^i + d\omega^k)$

$$\implies d\omega^k + \sum_i \omega_i^k \wedge \omega^i = 0$$

first Cartan's structure eq.

can think of these as the defining eqs. for ω_i^k

$$de_i = \sum_j \omega_i^j e_j$$

2.6.3 Cartan's second structure equation

Recall: $d(df) \equiv 0, \forall f$.

$$\begin{aligned} 0 &= d(de_i) = d\left(\sum_j \omega_i^j e_j\right) = \sum_j de_i \omega_i^j + e_j d\omega_i^j = \sum_j \left(\sum_k e_k \omega_j^k \wedge \omega_i^j + e_j d\omega_i^j\right) \\ &\implies \sum_j \omega_j^k \wedge \omega_i^j + d\omega_i^k = 0 \end{aligned} \quad (2.6.1)$$

Insight: We can study the

Different forms of ma express

1. skew-symmetric $3 \times 3 \longleftrightarrow$ Euclidean geometry
- 2.

2.7 Geometry of Surfaces

Similar to F-S, we will use frames to study the geometry of a surface. (change of the frame $\longleftrightarrow$)

Need frames that are adapted to the surface.

2.7.1 Def

A moving frame $e_1, e_2, e_3 : \mathcal{O} \rightarrow \mathbb{R}^3$ is called adapted to the surface $S \subseteq \mathbb{R}^3$. If e_3 is normal to S , i.e. $e_3 \cdot X = 0$. For all $p \in S \cap \mathcal{O}$.

In particular, if S is oriented with unit normal field n , then $n = e_3, \forall p \in S \cap \mathcal{O}$

e_1, e_2 are a basis of $T_p S, \forall p \in S \cap \mathcal{O}$

Ex: cylinder $\{x^2 + y^2 = \gamma^2\}$ $n(x, y) = x\partial_x + y\partial_y$

$\longrightarrow$ thus for example $e_1 = \partial_z, e_2 = e_1 \times e_3, e_3 = \frac{x\partial_x + y\partial_y}{\sqrt{x^2 + y^2}}$ is a frame on $\mathbb{R}^3 - \{0\}$ that is adapted to cylinder.

2.7.2 Lemma

Let S be an oriented surface with unit normal n , there exists an adapted frame on an open subset $\mathcal{U} \subseteq S$ iff there exists a non-moving tangent vector field V on $\mathcal{U}$

(e_1, e_2, e_3) adapted frame, so $v(p) = e_1(p)$ is

In particular, every parametric surface has an adapted frame. But, in general, a surface doesn't have an adapted frame which is defined on all of S .

Lemma: For every adapted frame e_1, e_2, e_3 , with dual frame $\omega^1, \omega^2, \omega^3$, then the element $dA = \pm \omega^1 \wedge \omega^2$ on $S \cap \mathcal{O}$.

Pf: Enough to

$$dA(e_1, e_2) = n \cdot (e_1 \times e_2) = e_3 \cdot (\pm e_3) = \pm 1$$

$$(\omega^1 \wedge \omega^2)(e_1, e_2) = 1 \cdot 1 - 0 \cdot 0 = 1$$

An adapted frame is called positively oriented if $dA = \omega^1 \wedge \omega^2 \iff n = e_1 \times e_2$.

If e_1, e_2, e_3 is an adapted frame on $\mathcal{O} \subseteq \mathbb{R}^3$, then

$$de_i(v) = \sum_{j=1}^3 \omega_i^j(v) e_j$$

$\forall p \in \mathcal{O}, v \in T_p \mathcal{O}$.

The restriction $(\omega^s)_i^j$ of ω_i^j to $p \in S \cap \mathcal{O}$ and $v \in T_p S$ gives a 1-form on $S \cap \mathcal{O}$.

Usually just write ω_i^j again instead of $(\omega^s)_i^j$

Let $n : S \rightarrow S^2$ be the Gauss map / unit normal field. Then, $\forall p \in S$

$$T_p n : T_p S \rightarrow \mathbb{R}^3 \quad (2.7.1)$$

Since $n \cdot n = 1$, $T_p n(v) \cdot n = 0$, $T_p(v) \in T_p S$

The

$$S_p = -T_p n : T_p S \rightarrow T_p S$$

is a linear map (Its also)

If e_1, e_2, e_3 is a adapted frame with forms ω_i^j .

For a curve curvature and torsion was telling you how the vector T and B are changing in the T -direction.

For a surface shape operator tells you how the normal vector changes in the two tangent directions.

Ex: plane $z = 0$ has constant normal field $n = \partial_z \implies S_p = 0$

Ex: Sphere of R has a Gauss map $n(p) = \frac{p}{R}$. So the directed derivatice is $T_p n(v) = \frac{v}{R}$

$$\implies S_p(v) = -\frac{v}{R}$$

There is a way to relate S to the acceleration of curves on S (similar to how the curvature of a curve used $\ddot{c}$)

Let $p \in S$, $v \in T_p S$ and curve $c : I \rightarrow S$, $c(0) = p$, $\dot{c} = v$

Does $\ddot{c}(0)$ depend really on the curve, or just on p and v ?

Really depend on c

2.7.3 Prop

If c is an aibit curve in S ,

$$\ddot{c}(t) \cdot n_{c(t)} = \dot{c}(t) \cdot S_{c(t)}(\dot{c}(t))$$

Proof: If c is a function in S , then $\dot{c}(t) \cdot n_{c(t)} = 0$, take $\frac{d}{dt}$

Thus all curves in S with a given velocity v at $p \in S$ have the same acceleration in the normal direction. That acceleration curves form rela

2.7.4 Def

The first fundamental form is

$$I_p : T_p S \times T_p S \rightarrow \mathbb{R} \quad (2.7.2)$$

def by $I_p(v, w) = v \cdot w$

The second fundamental form is

$$II_p : T_p S \times T_p S \rightarrow \mathbb{R} \quad (2.7.3)$$

def by $II_p(v, w) = v \cdot S_p(w) = I_p(v, S_p(w))$

Geometric meaning I is the Euclidean inner product of tangent vectors, encodes metric properties of S . For example, the length of a curve c on S is $L(c) = \int_a^b \sqrt{I_{c(t)}(\dot{c}(t), \dot{c}(t))} dt$

II will be used to define the curvature of S

If $\Phi : D \rightarrow S$ is a para of S , the local representation of I and II is def

$$I_{ij}^\Phi(x_1, x_2) = \frac{\partial \Phi}{\partial x_i} \cdot \frac{\partial \Phi}{\partial x_j}$$

$$II_{ij}^\Phi(x_1, x_j) =$$

Lemma: The first and second fundamental forms are symmetric

And equivalently I_{ij}^Φ and II_{ij}^Φ are symmetric in i and j .

Ex. Sphere with radius 1

$$\Phi(\theta, \varphi) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$$

$$\frac{\partial \Phi}{\partial \theta} = \dots$$

$$\frac{\partial \Phi}{\partial \varphi} = \dots$$

Last time

$$I_p(v, w) = v \cdot w$$

$$\Pi_p(v, w) = v \cdot \xi_p(w) = I_p(v, \xi_p(w))$$

3

Let $\Phi : D \rightarrow S\mathbb{R}^3$,

$$I_{ij}^\Phi(x_1, x_2) := I_{\Phi(x_1, x_2)} \left(\frac{\partial \Phi}{\partial x_i}, \frac{\partial \Phi}{\partial x_j} \right) = \frac{\partial \Phi}{\partial x_i} \cdot \frac{\partial \Phi}{\partial x_j}$$

$$\Pi_{ij}^\Phi(x_1, x_2) = \Pi_{\Phi(x_1, x_2)} \left(\frac{\partial \Phi}{\partial x_i}, \frac{\partial \Phi}{\partial x_j} \right) = -\frac{\partial \Phi}{\partial x_i} \Gamma_{\Phi(x_1, x_2)} n \left(\frac{\partial \Phi}{\partial x_j} \right)$$

Notice that $\frac{\partial \Phi}{\partial x_i} \cdot n = 0$,

$$\frac{\partial^2 \Phi}{\partial x_j \partial x_i}$$

Prop

Let Π_{ij} be the components of Π on adapted moving frame e_1, e_2, e_3 with connection forms ω_j^i and dual frame ω^3 , then we have $\omega_i^3 = \sum_{j=1,2} \Pi_{ij} \omega^j$, i.e. $\omega_1^3 = \Pi_{11} \omega^1 + \Pi_{12} \omega^2$, $\omega_2^3 = \Pi_{21} \omega^1 + \Pi_{22} \omega^2$

normal curvature in the u -direction

$$\kappa_p(u) = \Pi_p(u, u) = u \cdot \mathcal{S}(u)$$

Prop Any curve $c \in S$ which at a given point has the same speed has the same normal curvature at that point

$$\kappa_{c(t)}(\dot{c}(t)) = \dot{c}(t) \cdot \mathcal{S}_{c(t)}(\dot{c}(t))$$

Principle curvature of S at p if the maximum and minimum value of κ_p for $|u| = 1$, $u \in T_p S$

Find the extre of $u^T \mathcal{S}_p(u)$ under constant $u^T u = I$ $L = u^T \mathcal{S}_p u = -\lambda(u^T u - 1)$

Furthermore, if $\kappa_1 \neq \kappa_2$ ($u_1 \perp u_2$)

$$I_p = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Pi_p = \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix}$$

A point p s.t. $\kappa_1(p) = \kappa_2(p)$ if called an umbilical point (if $\kappa_1 = \kappa_2 = 0$, it's called planes) $\longrightarrow$ like a sphere
otherwise

1. $\kappa_1 \cdot \kappa_2 > 0$, is called elliptic point (ellipotoid)
2. $\kappa_1 \cdot \kappa_2 < 0$ is called hyperbolic point (hyperblend)
3. $\kappa_1 \cdot \kappa_2 = 0$ is called parabolic point (cylinder)

3.1 Darboux frame

A moving frame on $\mathbb{R}^3$ adapted to S if called Darboux frame if e_1 and e_2 are the principal direction.

A unit speed curve is called a line of curvature if $\dot{c}$ is always a principal direction.

Prop For a Darboux frame

$$d\kappa_1(e_2) = (\kappa_1 - \kappa_2)\omega_1^2(e_1)$$

$$d\kappa_2(e_1) = (\kappa_1 - \kappa_2)\omega_1^2(e_2)$$

$$\sum_k \omega_k^i \wedge \omega^k + d\omega^i = 0$$

$$\sum_j \omega_j^k \wedge \omega_j^j + \omega_i^k = 0$$

$$H = \frac{1}{2}(\kappa_1 + \kappa_2) = \frac{1}{2} \text{tr}(\text{II})$$

1. $d\omega_1^2 = -\kappa\omega^1 \wedge \omega^2$
2. $\omega_1 \wedge \omega^2 + \omega^1 \wedge \omega_2^3 = 2H\omega^1 \wedge \omega^2$
- 3.

$$\begin{aligned} d\omega_1^2 &= \omega_1^3 \wedge \omega_3^2 = \\ &= \det(\text{II})\omega^1 \wedge \omega^2 = \kappa\omega^1 \wedge \omega^2 \end{aligned} \tag{3.1.1}$$

Let $F : S \rightarrow \bar{S}$

pushforward of curves

$$c : I \rightarrow S \quad F_*c : I \rightarrow \bar{S}$$

pushforward of tangent vectors

$$v \in T_p S \quad c(t) : I \rightarrow S \text{ s.t. } c(0) = v$$

$$F_*v = w = (F \circ c)'(0) = T_p F(\dot{c}(0))$$

pullback of functions:

$$f : \bar{S} \rightarrow \mathbb{R} \quad F^*f = f \circ F : S \rightarrow \mathbb{R}$$

pullback of 1-forms: let α be a 1-form on $\bar{S}$, $(F^*\alpha)_p(v) = \alpha(T_p F(v))$

pullback of 2-forms: $(F^*\beta)_p(v_1, v_2) = \beta(T_p F(v_1), T_p F(v_2))$

$$F^*(\alpha \wedge \beta) = (F^*\alpha) \wedge (F^*\beta)$$

$$F^*(d\alpha) = d(F^*\alpha)$$

$\bar{\alpha}, \bar{\beta}$ are on $\bar{S}$

$$F^*(\bar{\alpha} \wedge \bar{\beta})_p(u, v) = (\bar{\alpha} \wedge \bar{\beta})_{F(p)}(T_p F(u), T_p F(v)) = \dots = (F^*\bar{\alpha}) \wedge (F^*\bar{\beta})_p(u, v)$$

Let $\Phi : D \rightarrow S$ be a chart on S at p , then $F \circ \Phi : D \rightarrow \bar{S}$ at $F(p)$

$$(dF^*\bar{\alpha})_{\Phi(x,y)}(T_{(x,y)}\Phi(\partial_x), T_{(x,y)}\Phi(\partial_y)) = \frac{\partial}{\partial x} - \frac{\partial}{\partial y} \quad (3.1.2)$$

lalign

An isometry $F : S \rightarrow \bar{S}$ is a bijection preserving the inner product $(T_p F)u \cdot (T_p F)v = u \cdot v \quad \forall u, v \in T_p S$

A local isometry $F : S \rightarrow \bar{S}$ is a smooth map that is locally an isometry i.e. $\forall p \in S$ open neighborhood U s.t. $F|_U$ is an isometry.

The distance between $p, q \in S$ is $d(p, q) = \inf\{L(c) \mid c : I \rightarrow S, c(0) = p, c(1) = q\}$

Isometries preserve distance i.e. $F : S \rightarrow \bar{S}$ is an isometry, then $d(p, q) = d(F(p), F(q))$

3.1.1 Gauss's Theorema Egregium

The Gaussian curvature of a surface is invariant under local isometries i.e. $F : S \rightarrow \bar{S}$ local isometry $\kappa(p) = \bar{\kappa}(F(p))$

3.1.2 Lemma

If $\{e_1, e_2, e_3\}$ is a moving frame on $\mathbb{R}^3$ adapted to S , then $\{e_1(p), e_2(p)\}$ is an ONB

3.1.3 Lemma

Let $F : S \rightarrow \bar{S}$ be an isometry, let $\{e_1, e_2\}$ be tangent frame on S with dual frame $\{\omega^1, \omega^2\}$ and connection form ω_1^2 , then

- 1.
- 2.
- 3.

Let S be a connected surface s.t. every point is umbilic, then S has constant Gaussian curvature $\kappa \geq 0$, and

1. if $\kappa = 0$, S is point of plane
2. if $\kappa > 0$, S is point of a sphere with radius $\frac{1}{\sqrt{\kappa}}$

Pf $\kappa_1 = \kappa_2 = \lambda$

1. if $\kappa = 0$, then $\lambda = 0$, $\mathcal{S} = 0$, n_p is
- 2.

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Assume $\kappa_1 > \kappa_2$

$$\begin{aligned}\omega_1^2 &= \underbrace{\omega_1^2(e_1)}_{f_1} \omega^1 + \underbrace{\omega_1^2(e_2)}_{f_2} \omega^2 \\ d\omega_1^2 &= df \wedge \omega^1 + f_1 d\omega^1 + f_2 d\omega^2 \\ df_1 &= df_1(e_1)\omega^1 + df_1(e_2)\omega^2 \\ df_2 &= df_2(e_1)\omega^1 + df_2(e_2)\omega^2 \\ d\omega^1 &= \omega_1^2 \wedge \omega^2 = (f_1\omega^1 + f_2\omega^2) \wedge \omega^2 = f_1\omega^1 \wedge \omega^2 \\ d\omega^2 &= \dots = -f_2\omega^2 \wedge \omega^1 \\ d\omega_1^2 &= df_1(e_1)\omega^2 \wedge \omega^1 + (f_1)^2\omega^1 \wedge \omega^2 - (f_2)^2\omega^2 \wedge \omega^1 = \\ \kappa &= df(e_2) - (f_1)^2 - df_2(e_1) + (f_2)^2 \leq 0\end{aligned}$$

For a curve $c(t)$ with constant speed on a plane, the following are equivalent

1. c is a straight line
2. $\ddot{c}(t) = 0$
3. $\dot{c}(t)$ are para, $\forall t \in [0, 1]$
4. c is the shortest connection between $c(0)$ & $c(t)$

Let $\{e_1, e_2, e_3\}$ be adapted to $c(t)$ s.t. $e_1(c(t)) = \frac{\dot{c}(t)}{v}$. Since $\ddot{c} \cdot c = 0$, $\exists \kappa_g, \kappa_n$ s.t.

$$\frac{\ddot{c}(t)}{v^2} = \kappa_g(t)e_2(c(t)) + \kappa_n(t)e_3(c(t))$$

$$\kappa(t) = \frac{\|\ddot{c}(t)\|}{v^2} = \sqrt{\kappa_g^2 + \kappa_n^2}$$

on the other hand

$$\dots$$

$$\begin{cases} \kappa_g = \omega_1^2 \left(\frac{\dot{c}(t)}{v} \right) \\ \kappa_n(t) = \omega_1^3 \left(\frac{\dot{c}(t)}{v} \right) \end{cases}$$

A constant speed curve c with $\dot{c}(t) = e_1(c(t))$ on a surface is called a geodesic if

1. $\ddot{c}(t)$ is in normal direction
2. $\kappa_g = 0$
3. $\omega_1^2(\dot{c}(t)) = 0$

Def Let $c : I \rightarrow S$ ve a smooth curve. A vector field along c is a smooth map $X : I \rightarrow \mathbb{R}^3$ s.t. $X(t) \in T_{c(t)}S$, $\forall t \in I$.

$$X = f_1 e_1(c(t)) + f_2 e_2(c(t))$$

$$\frac{d}{dt}X = \frac{\nabla}{dt}X + \Pi_{c(t)}(\dot{c}(t), X(t)) \cdot n_{c(t)}$$

The covariant derivative has the following properties

1. Leibniz

$$\frac{\nabla}{dt}(\lambda X) = \dot{\lambda}X + \lambda \frac{\nabla}{dt}X$$

2. Chain rule

$$\frac{\nabla}{ds}(X \circ \phi)(s) = \dot{\phi}(s) \frac{\nabla}{ds}X(\phi(s))$$

3. For every X, Y along c , we have

- 4.

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Every point p has a sufficiently small neighborhood U , s.t. $\forall p \in U$, there is exactly one

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Gauss-Bonnet formula for compact oriented surface

$$\frac{1}{2\pi} \int_S \kappa \, dA = 1(\text{number of holes}) = \chi(S)$$

Eg: sphere with radius R

$$\frac{1}{2\pi} \int_S \frac{1}{R^2} \, dA = \frac{1}{2\pi R^2} 4\pi R^2 = 2$$

κ changes under def; κ depends on I , local geometry

